

Ghost freedom does not constrain the sound speed of a shift-symmetric condensate

Carl Zimmerman¹

¹*Briar Creek Tech, Charlotte, North Carolina, USA*

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For a shift-symmetric scalar with Lagrangian $K(\mathcal{Q})$, where $\mathcal{Q}$ is the charge variable conjugate to the shift symmetry, we note the identity $K'' = n/(\mathcal{Q}c_s^2)$ with $n = K'$ the conserved charge density. Because $n > 0$ and $\mathcal{Q} > 0$ on any physical branch, this shows that the no-ghost condition $K'' > 0$ is *equivalent* to $c_s^2 > 0$ and therefore constrains only the **sign** of the sound speed—never its scaling with charge density, and never its magnitude. This contradicts a natural and, in our experience, easily made assumption: that stability forces the sound speed of such a condensate to redshift as $c_s^2 \propto a^{-3}$, driving it to dust. That behaviour is the special case in which K' possesses a *simple interior zero*. We exhibit an explicit ghost-free counterexample, $K'(\mathcal{Q}) = n_1[1 + (s/c_1)\ln(\mathcal{Q}_1/\mathcal{Q})]^{-1/s}$, whose sound speed *rises* as the charge dilutes, $c_s^2 = c_1(n/n_1)^{-s} \propto a^{+3s}$. Here $K' > 0$ for all $\mathcal{Q} > 0$ and vanishes only at the domain boundary, logarithmically: 298 decades of charge buy 6 decades of K' . Numerically, $K'' > 0$ at 400 sampled charges spanning 45 decades in n with none skipped, minimum $K'' = 4.9 \times 10^{-27}$, and the closed-form sound speed is reproduced to 2×10^{-52} at 60-digit precision. The equation-of-state fixed point $w = -1$ survives, relocated to the boundary of the charge domain. The consequence is that warm shift-symmetric dark sectors are not excluded by stability, and any argument that closes them must appeal to something other than ghost freedom. We state plainly what this result does *not* do: it removes an obstruction, it does not build a viable cosmology, and one class of candidate application is reported here as failing three independent observational gates.

I. SETUP

Consider a scalar ϕ with shift symmetry $\phi \rightarrow \phi + \text{const}$, in a configuration $\phi = \mathcal{Q}_0 t + \psi(\mathbf{x})$, so that the time component of the gradient plays the role of a chemical potential. Write the Lagrangian as $K(\mathcal{Q})$. Then

$$n = K'(\mathcal{Q}), \quad \rho = \mathcal{Q}K' - K, \quad p = K, \quad (1)$$

with n the conserved Noether charge density of the shift symmetry. These are the standard thermodynamic relations of a charged condensate, with $\mathcal{Q}$ the chemical potential conjugate to n ; they follow either by direct variation or from the first law $d\rho/dn = (\rho + p)/n$.

The sound speed and the no-ghost condition are

$$c_s^2 = \frac{dp}{d\rho} = \frac{K'}{\mathcal{Q}K''}, \quad \text{no ghost} \iff K'' > 0. \quad (2)$$

II. THE THEOREM

Rearranging Eq. (2) gives the identity

$$\boxed{K'' = \frac{n}{\mathcal{Q}c_s^2}} \quad (3)$$

On any physical branch $n > 0$ and $\mathcal{Q} > 0$, so the sign of K'' is the sign of c_s^2 and nothing else. Hence:

Theorem. For a shift-symmetric condensate, ghost freedom is *equivalent* to $c_s^2 > 0$. It places no constraint whatsoever on the magnitude of c_s^2 , nor on its scaling with n (equivalently, with the scale factor through $n \propto a^{-3}$).

The theorem is trivial once Eq. (3) is written down, which is precisely why the assumption it refutes is easy to make. The tempting argument runs: K' must vanish somewhere for the charge to dilute; near a zero of K' one has $K'/K'' \sim (\mathcal{Q} - \mathcal{Q}_*)$, hence $c_s^2 \propto n \propto a^{-3}$; therefore any ghost-free condensate cools to dust. Every step is correct *given a simple interior zero*. The conclusion does not follow in general, because that hypothesis is an extra assumption and not a consequence of stability.

III. THE COUNTEREXAMPLE

Let

$$K'(\mathcal{Q}) = n_1 \left[1 + \frac{s}{c_1} \ln \frac{\mathcal{Q}_1}{\mathcal{Q}} \right]^{-1/s}, \quad s > 0, \quad (4)$$

which integrates the ODE $d \ln \mathcal{Q} / d \ln n = c_s^2$ with $c_s^2 = c_1(n/n_1)^{-s}$. Then

$$c_s^2 = c_1 \left(\frac{n}{n_1} \right)^{-s} \propto a^{+3s}, \quad (5)$$

so the sound speed *rises* as the charge dilutes: the sector is cold when dense and warm when rarefied.

Three properties make this a genuine counterexample rather than a formal one.

a. It is ghost-free everywhere on the branch. $K' > 0$ for every $\mathcal{Q} > 0$, and $K' \rightarrow 0$ only at the domain boundary $\mathcal{Q} \rightarrow 0$ —logarithmically, so the approach is extraordinarily slow: 298 decades of $\mathcal{Q}$ buy 6 decades of K' . There is no interior zero, which is exactly the hypothesis the tempting argument requires.

b. It is verified, not merely asserted. Independent reconstruction from the defining ODE (rather than by substitution into the closed form) gives $K'' > 0$ at 400 sampled charges spanning 45 decades in n , none skipped, with minimum $K'' = 4.87 \times 10^{-27}$; and c_s^2 from Eq. (2) reproduces $c_1(n/n_1)^{-s}$ to a relative 1.9×10^{-52} at 60-digit precision.

c. The dark-energy fixed point survives. $K(Q_1) - K(0)$ is finite and analytically bounded by $n_1 Q_1$, so the $w = -1$ stationary point is preserved—relocated to the boundary of the charge domain rather than destroyed. A condensate of this type can still supply an exact cosmological-constant branch.

d. A family that does not work, for the record. It is worth recording which natural family fails, since we tried it first. Taking $K' \propto (Q - Q_*)^m$ gives $c_s^2 = (Q - Q_*)/(mQ) \propto n^{1/m} \propto a^{-3/m}$, which declines for every $m > 0$; and $K'' > 0$ requires $m > 0$, so $m < 0$ is a ghost rather than an escape. The rising branch is not contained in that family at all; the map to Eq. (4) is $s = -1/m$.

IV. WHAT FOLLOWS, AND WHAT DOES NOT

a. What follows. Arguments that close warm shift-symmetric dark sectors on stability grounds are invalid as stated. In particular, a chain of the form “the excitation is ghost-free, therefore $c_s^2 \propto a^{-3}$, therefore it is dust, therefore it clusters and cannot be kept smooth” fails at its second step. Any such closure must instead be argued from observation—the cosmic microwave background, structure growth, or the phase-space constraints—and those arguments have different and generally weaker leverage.

b. What does not follow. Removing an obstruction is not the same as building a theory, and we decline to present it as one. In the application that motivated this work—a condensate supplying the cosmological dark sector while remaining smooth enough in galaxies not to double-count a modified-gravity phantom—the warm branch was constructed and then failed three indepen-

dent observational gates under adversarial review: the tensor speed constraint from GW170817, cluster scales at the low end of the charge window, and the radial acceleration relation, which degrades from 0.108 to 0.113 dex through a coherent deep-MOND tilt. A window on the required sound speed remained short by a factor of 1.28, which is a small number but not zero. We report this because a result that removes an obstruction is easy to oversell, and the honest scope of the present paper is the theorem and its counterexample, not the application.

c. A methodological note. The assumption refuted here survived unchallenged in our own work for an extended period and was used to close a live line of investigation. It failed for a specific and recognisable reason: a correct formula—the near-a-simple-zero expansion—was generalised beyond the regime in which it holds. We record this because the same error mode accounted for the majority of incorrect exclusions we have had to withdraw, and because a stability argument is exactly the kind of claim that is rarely revisited once made.

V. REPRODUCIBILITY

The construction, the 45-decade ghost-freedom scan, the 60-digit sound-speed check and the $w = -1$ boundary argument are implemented in the accompanying scripts, each of which exits zero. The counterexample was independently rebuilt from the defining ODE by a separate analysis that did not import the closed form, and that reconstruction is included.

ACKNOWLEDGMENTS

The k-essence sound-speed relation and the no-ghost condition are standard; what is claimed here is the observation that they are equivalent statements for a shift-symmetric condensate, and the explicit rising- c_s^2 counterexample. The analysis, scripts and adversarial verification were carried out with Claude (Anthropic).